

Assessing the effect of pan versus isoform specific HDAC inhibition in two models of achromatopsia

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Background

Histone deacetylase (HDAC) overactivity has been shown in the photoreceptors of several models of inherited retinal disease (IRD) and has been linked to the death of photoreceptors. Administration of pan-HDAC inhibitors was shown to reduce both rod and cone degeneration in models of retinitis pigmentosa (RP) and in the *Pde6c^{cpfl1}* model of achromatopsia, but the role of specific HDAC isoforms is less understood. HDAC6 inhibition has previously shown potential as a neuroprotective target, providing improvements in visual function and cell morphology in a zebrafish model of inherited blindness, and conferring increased cone survival in an *ex vivo* model of RP. However, the role of HDAC6 inhibition has yet to be investigated in models of cone-only degeneration, such as achromatopsia.

Methods

We tested pan- and isoform-specific HDAC inhibitors in two murine models of achromatopsia carrying mutations in the *Gnat2* or *Pde6c* gene, while also expressing GFP exclusively in cones (referred to as *Gnat2.GFP* or *Pde6c.GFP* herein). We intravitreally injected the pan-HDAC inhibitor SAHA or the HDAC6 inhibitor, Tubastatin A, at 12 or 20 days of age (P12 or P20). Eyes were collected four days or twelve days after treatment for histological and molecular analyses.

Results

In *Pde6c.GFP* mice, SAHA treatment administered at P12 caused an 18% increase in cone numbers compared to controls when eyes were assessed four days after treatment ($n=7$, $P=0.0015$). However, when eyes were collected twelve days after treatment, there were no protective effects noted, suggesting a lack of sustained protection ($n=6-7$, $P>0.05$). Similarly, in *Gnat2.GFP* mice that were treated with Tubastatin A at P20, retinæ that were collected four days after treatment showed an 11% improvement in cone numbers compared to controls ($n=7$, $P<0.0001$), with this effect lost when eyes were collected twelve days post-treatment ($n=5-6$, $P>0.05$).

Conclusion

While we noted significant neuroprotection after administration of SAHA in *Pde6c.GFP* mice and Tubastatin A in *Gnat2.GFP* mice, these protective effects were only noted when retinæ were assessed four days after treatment but not twelve. As both drugs do not appear to have sustained treatment effect, the next steps in this study involves implementing an *ex vivo* system to provide sustained delivery of treatments which will allow us to understand whether long term protection of the cone photoreceptors occurs and assess long term toxicity of treatment application.